$L[f'(t)] = sF(s) - f(0)$

$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$

However, poles < 0

Const input

how much things settle once dynamics settle

DC Gain: Transfer fct $G(0)$

System gain: $\frac{10(s+4)}{s+a}$ → is system gain

Steady State means what the output becomes after the transients die out.

$y(t) \xrightarrow[t\to\infty]{} y_{ss}$

Fourier: What frequencies are present

Laplace: How system evolves over time

- Region of Convergence: Which exponential probes does this signal behave nicely.
- $\int_0^\infty f(t)e^{-st}dt = F(s)$ — if e^{st}, $\int e^{st} \to \infty$
- $a_n\frac{d^n x}{dt^n} + \cdots + a_1\frac{dx}{dt} + a_0 x = u(t)$ (LTI)

  $x(t) = \sum e^{(\sigma_i + j\omega_i)t} \cdot (\text{consts})$
- Poles, where system diverges.
- If your system were spring, poles are the resonant frequencies where the spring wants to oscillate or grow naturally
- For rational $F(s) = \frac{N(s)}{D(s)}$ $\quad s = \sigma + i\omega$
  - Cauchy's residue Theorem: $\int_{\sigma - i\infty}^{\sigma + i\infty} F(s)e^{st}ds = 2\pi i \sum \text{residues at poles} \cdot e^{s_i t}$

    $\text{Res}[F(s), a] = \lim_{s\to s_i}(s - s_i)F(s)$
  - So Inverse Laplace: $\frac{1}{2\pi i}\int_{\sigma - i\infty}^{\sigma + i\infty} F(s)e^{st}ds = f(t)$

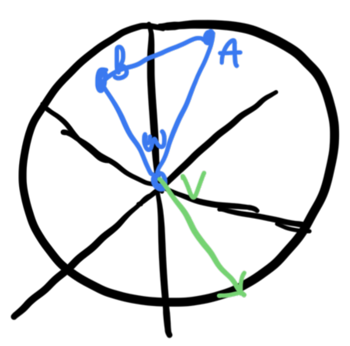
Root Locus: how poles of a system move in a complex plane as you vary some parameter – Controller gain K

$1 + KG(s) = 0$ or $D(s) + KN(s) = 0$

- Tracking modes of system as you adjust gain
  - If $G(s) = \frac{1}{s(s+2)}$ $\quad 1 + KG(s) = 0 \to s(s+2) + K = 0$

    $s = -1 \pm \sqrt{1-K}$

Quaternions:



$q = (w + \vec{v})$ is quaternion

instead of using matrix for rotation, use quaternion.

- Makes chaining easier too.

Transfer Functions

Output = (Something) · input

Transfer function: $G(s) = \frac{Y(s)}{U(s)}$

- $Y(s)$ = Laplace of the output $y(t)$
- $U(s)$ = Laplace transform of input $u(t)$
- $G(s)$ = fct of s that tells you how the system responds to an input in Laplace Domain

Open-system: $Y(s) = G(s)U(s)$

Closed-system: $U(s) = C(s)(R(s) - Y(s))$

- $R(s)$ = Laplace of reference $\quad C(s)$ = Controller

$Y(s) = G(s)U(s)$ — Solve for $\frac{Y(s)}{U(s)}$

$T(s) = \frac{Y(s)}{U(s)} = \frac{G(s)C(s)}{1 + G(s)C(s)}$

PID: Proportional + Integral + Derivative

$U(t) = K_p e(t) + K_I \int e(t)dt + K_d \frac{de}{dt}$

$e(t) = r(t) - y(t)$

$r(t)$ = reference

$y(t)$ = output

$u(t)$ = action

don't know, only know $y(t), r(t)$

$r(t) \to$ [Controller] (PID) $\xrightarrow{u(t)}$ [System] ($y(t) = G(s)u(t)$) $\to (y(t))$ — feedback

Say you want $r(t)$, if sensor say $y(t)$ and $r(t) \neq y(t)$, throttle adjusts

LQR: State Space optimal controller for linear systems

$\dot{x}(t) = Ax(t) + Bu(t)$

- $x(t)$ = state vector
- $u(t)$ = control input
- A, B = System matrices

- LQR chooses $u(t)$ to minimize a cost fct

  $J = \int_0^\infty (x^TQx + u^TRu)dx$ — if desired state $\neq 0$, $\tilde{x} = (x(t) - x_{des})$ then $\tilde{x}^TQ\tilde{x}$ & $u(t) = -K\tilde{x}(t)$

  Q = penalty on state deviations

  R = penalty on control effort

How LQR produces a controller

- $u(t) = -Kx(t)$: Solu gives State Feedback
  - K = matrix gain computed by solving the Ricatti eq
    - $A^TP + PA - PBR^{-1}B^TP + Q = 0$
    - P = Solu to Ricatti eq
    - $K = R^{-1}B^TP$

$\dot{x} = (A - BK)x$ (Closed-loop) → Eigenvales = closed loop poles

State-Space Model General Form

$\dot{\vec{x}} = A\vec{x} + B\vec{u}$

$\vec{y} = C\vec{x} + D\vec{u}$

$\vec{x}$ = State vectors (vars describing system)

$\vec{u}$ = (external force or control)

$\vec{y}$ = output

A = how states affect each other

B = how input affects states

C = which states are measured

D = direct input to output effect

Fixed points: $\dot{\vec{x}} = \vec{f}(\vec{x})$, when $\vec{f}(\vec{x}^*) = 0$

Ex: $x' = x(x-1)$ fixed points $x = 0, 1$

Stability: $f'(x^*) < $ or > 0. Higher dim: $J = \frac{\partial \vec{f}}{\partial \vec{x}}$ & Check eigenvals

Bifurcations: $\dot{\vec{x}} = \vec{f}(\vec{x}, \vec{r})$, parameter $\vec{r}$.

- Fixed points appear or dissapear
- Stability changes

$x' = r - x^2$ Fixed points $x = \pm\sqrt{r}$

$r < 0$: no real fixed point

$r = 0$: one degen case

$r > 0$: two fixed points (one stable, one not)

Hessian & Jacobian Shit are diff.

- Prob B happens, then A given B
- Prob A happens, then B
  - Describes $P(A \cap B)$

A: Bayes Rule: $P(\theta|y) = (P(y|\theta)P(\theta)) / P(y)$ — Posterior; likelihood; prior; marginalization

Frequency domain Analysis: From wave eq: $f(x - ct)$ is general solution

$f(x) = \sum_{-\infty}^{\infty} A(k)e^{ikx} = \int A(k)e^{ikx}dk$

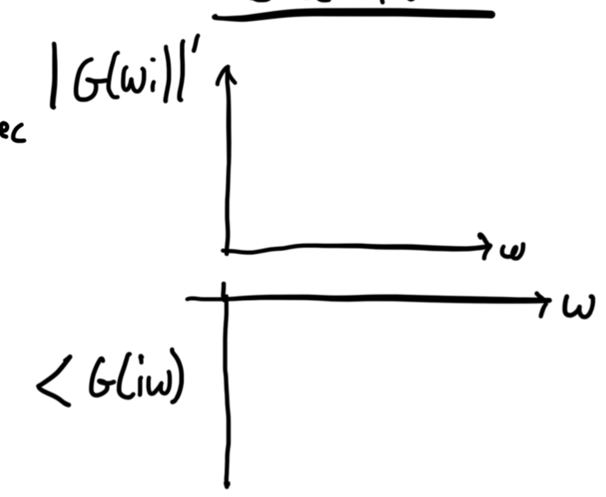
$|G(i\omega)| = (Re[G(i\omega)]^2 + Im(G(i\omega))^2)^{\frac{1}{2}}$

$\theta = \angle G(i\omega)$ $\quad a + ib$

$= \arctan 2(b, a)$

$u(t) = A\sin(\omega t) \to$ [LTI, $G(s)$] $\to y_{ss}(t) = A|G(i\omega)|\sin(\omega t + \theta)$

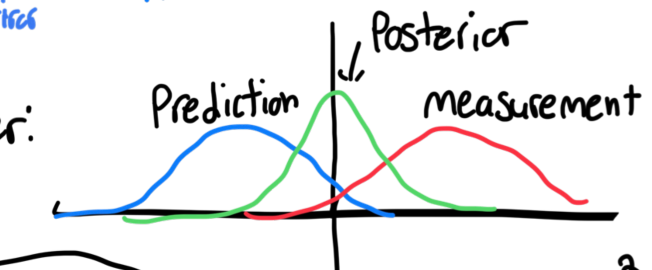
Bode Plot



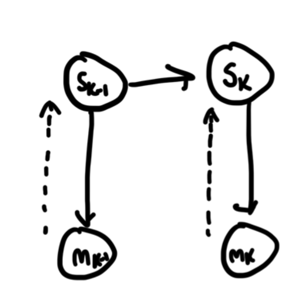
Hertz: $\frac{1}{T}$:= cycles per time step $\quad w = 4\pi$ rad/sec

$f = \frac{4\pi}{2\pi} = 2$ Hz

Kalman Filter:



Kalman Gain: $\hat{x} = x_{pred} + K(z - x_{pred})$ $\quad K = \frac{\sigma^2_{pred}}{\sigma^2_{pred} + \sigma^2_{meas}}$



$P(s_k|m_k) = \frac{P(m_k|s_k) \cdot P(s_k|m_{k-1})}{N}$ → how to get this? (Prior) : Predict — Bayesian Filter

Chapman-Kolmogorov: $P(s_k|m_{k-1}) = \int P(s_k|s_{k-1})P(s_{k-1}|m_{k-1})ds_{k-1}$ — Update

$P(s_k|m_{k-1}) = \mathcal{N}(\hat{\mu}_k, \hat{P}_k)$

$\mu_k = A_{k-1}\mu_{k-1}$

$\hat{P}_k = A_{k-1}P_{k-1}A^T_{k-1} + Q_{k-1}$

$P(s_k|m_k) = \mathcal{N}(\mu_k, P_k)$

$\mu_k = \hat{\mu}_k + K_k(m_k - H_k\hat{\mu}_k)$

$P_k = (I - K_kH_k)\hat{P}_k$

$K_k = \hat{P}_k H_k^T(H_k\hat{P}_kH_k^T + R_k)^{-1}$

$\vec{s}_k = A_{k-1}\vec{s}_{k-1} + \vec{q}_{k-1}$ (transition matrix; process noise) $\quad q_{k-1} \sim \mathcal{N}(0, Q_{k-1})$

$\vec{m}_k = H_{k-1}\vec{s}_k + \vec{r}_k \quad r_k \sim \mathcal{N}(0, R_k)$

$P(\vec{s}|\vec{s}_{k-1}) = \mathcal{N}(A_{k-1}\vec{s}_{k-1}, Q_{k-1})$

$P(\vec{m}|\vec{s}_k) = \mathcal{N}(H_k s_k, R_k)$

Simplifies to:

Plug into Bayesian Filter